

Evaluation of lower leg muscle activities during human walking assisted by an ankle exoskeleton

Wei Wang, Jianyu Chen, Yandong Ji, Wei Jin, Jingtai Liu, *Member, IEEE* and Juanjuan Zhang

Abstract—Wearable robots like ankle exoskeletons have demonstrated the capability to enhance human mobility and to reduce biological efforts of human locomotion. The type of assistance provided by ankle exoskeletons could influence the lower leg muscle activities during human walking. This paper aimed to systematically evaluate the lower leg muscle activities under different ankle exoskeleton assistance conditions. We measured multiple electromyography-based metrics of five lower leg muscles while the participants walked with an ankle exoskeleton on a treadmill. Nine assistance conditions which combined three peak times (46, 49 and 52% of stride time) and three peak torque levels (0.3, 0.5 and 0.7 N·m·kg⁻¹) were applied to assist plantarflexion during ankle push-off. Nine healthy subjects participated in the experiments. Of all investigated muscles, the activity level of lateral soleus was influenced the most when exoskeleton assistance was applied. The root-mean-square of lateral soleus activity reduced by 33.6 ± 14.0% under one assistance condition compared to walking without the exoskeleton. Our results can be used to guide studies on mechanical and control designs to improve neuromuscular interactions between exoskeletons and wearers.

Index Terms—Ankle exoskeleton, wearable robots, gait assistance, plantarflexion, muscle activity, electromyography

I. INTRODUCTION

In the past few decades, many exoskeletons have been designed to enhance human locomotion performance and to assist motion rehabilitation [1]–[3]. A fairly small number of these devices could reduce biological efforts of walking, running or climbing stairs with predefined assistance patterns [4]–[6]. Recent advances in human-in-the-loop (HIL) optimization have demonstrated great potential to improve wearable assistive robotics performance [7]–[10]. It uses real-time measurements of human physiological signals as optimization objectives to continuously update exoskeleton control parameters and thus to generate customized assistance patterns. In this process, subjects experience a wide range of assistance conditions.

In previous studies, metabolic cost was widely used as the HIL optimization objective of exoskeleton assistance conditions. Multiple studies have discussed the metabolic cost of human beings during exoskeleton assisted walking [4],

[5], [11]–[16]. They found that when appropriate exoskeleton assistance patterns were applied, the metabolic rate of the human body could be reduced during walking. Metabolic rate has been commonly used as a physiological indicator for the effectiveness of various exoskeleton assistance patterns, and thus it often serves as the objective of HIL optimization [7]–[9]. However, measuring of metabolic cost, which is usually through an indirect calorimetry, is noisy, delayed and sparsely sampled. These characteristics lead to long evaluation times and thus inefficient optimization process. For example, one study optimized four lower limb exoskeleton control parameters based on metabolic cost within 64 minutes walking [8]. Another one optimized two parameters in an average of 21.4 ± 1.0 minutes [9]. Meanwhile, both indirect calorimetry equipment and non-optimized exoskeletons are uncomfortable to wear and may induce faster human fatigue. This results in unpleasant experimental experiences and inaccurate experimental outcomes. Therefore, there have been motivations to explore the possibility of using other physiological responses as HIL optimization objectives [8]. One such reasonable candidate is muscle activity. Currently, the effectiveness of muscle activity levels in being used as optimization objectives has not been thoroughly evaluated. A key measurement method for muscle activity intensity is the surface electromyography (EMG). EMG signals contain important information about energy consumption of muscle contractions and have been widely used to control wearable robotic devices and to evaluate their effects on human performance [17]–[20]. It is necessary and important to systematically evaluate relevant muscle activities in different ankle exoskeleton assistance conditions to provide comprehensive understanding about the mechanism of human walking assisted by an ankle exoskeleton. It is also important to verify the effectiveness of using muscle activities in HIL optimization.

Our goal was to systematically investigate the influence of assistance conditions on lower leg muscle activities while human walking with an ankle exoskeleton. We focused mainly on lower leg muscles which were closer to the assisted human joint and have been shown to be more effected during exoskeleton assisted human walking [12], [21], [22]. We used a unilateral ankle exoskeleton to assist plantarflexion movements during human walking and changed assistive torque profiles in a broad range to study the influence of different exoskeleton assistance patterns on the exoskeleton-side lower leg muscle activities of walking. We expected our study to help to evaluate exoskeleton performance and develop advanced control algorithms based on EMG. Furthermore, results of this study could help to guide the selection of HIL optimization objective.

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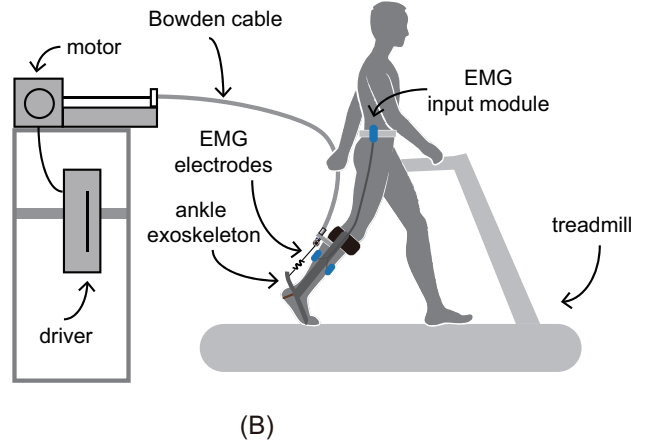
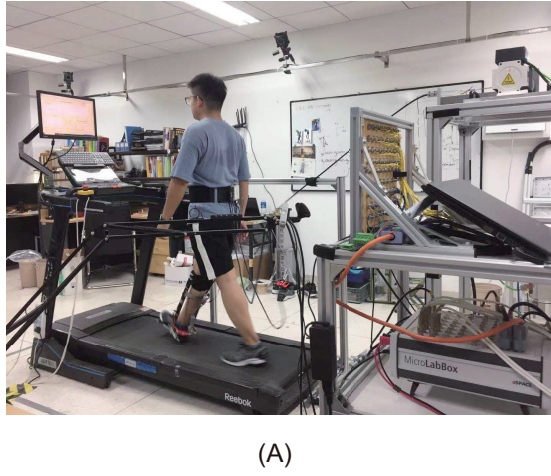


Fig. 1. **Ankle exoskeleton experimental platform.** (A) Photograph of the experimental setup. (B) Schematic diagram of the experimental setup. Subjects walked on a treadmill while wearing an ankle exoskeleton on right ankle joint. The ankle plantarflexion assistive torque generated from a motor and transmitted to the ankle exoskeleton through a flexible Bowden cable. EMG electrodes were placed on the right lower leg to measure muscle activities.

II. METHODS

We investigated the lower leg muscle activities on a tethered unilateral ankle plantarflexion exoskeleton testbed. All subjects walked on a treadmill under a fixed speed while wearing the exoskeleton on the right foot. The assistive torque was enforced with a combination of proportional-derivative control and iterative learning control. We applied nine different assistive torque profiles, each defined by a unique combination of peak time and peak torque values, and collected the corresponding EMG measurements of five lower leg muscles. We calculated three metrics from the raw EMG signals and performed statistical analysis on them among various walking conditions.

A. Ankle Exoskeleton Experimental Platform

We built an ankle exoskeleton assisted human walking experimental platform (Fig. 1). It consisted of an ankle exoskeleton that interfaces with the human shank and foot, a high-speed control system responsible for torque control, multiple sensors, and an off-board actuation system that generated and transmitted assistive torque to the exoskeleton. Hardware configuration of the system is shown in Fig. 2(B).

Ankle exoskeleton

The ankle exoskeleton mainly comprised of three components: a shank frame, a foot frame and an titanium ankle lever, joined together with two joint shafts on either side of the human ankle (Fig. 2(A)) [23]. Assistive torque was transmitted to the titanium lever through a Bowden cable. The cable was threaded with a pulley and connected with a series spring which increased the passive compliance of human exoskeleton interactions [24]. A heel rope was placed in the bottom of a shoe that was used to lift the heel when assistive torque was provided.

Control system

A real-time controller (DS1202, dSPACE, Paderborn, GmbH) was used to sample measured data from sensors and to generate motor velocity commands. Motor velocity commands were generated at a frequency of 500 Hz.

Sensors

Surface electromyography (sEMG) signals were measured by a wired EMG system (Bagnoli, Delsys, MA, USA). The assistive torque was measured by a Wheatstone-bridge consisting of four strain gauges (KFH-6-350-C1-11L1M2R, OMEGA Engineering, CT, USA). Strain gauges signals were then amplified by a signal conditioner (IAA100, Futek, CA, USA). Two digital optical encoders (E5, US Digital, WA, USA) were installed on motor shaft and exoskeleton joint shaft respectively to measure motor position and ankle joint angle. A footswitch (McMaster-Carr, Aurora, OH, USA) was attached inside the exoskeleton shoe to detect heel strike.

Actuation system

The actuation system was composed of an AC brushless servo motor, a 5:1 planetary gear, and a motor driver (BSM90N-175AA, GBSM90-MRP120-5 and MF180-04AN-16A, ABB, Zurich, Swiss). Then the torque generated from the motor was transmitted to the ankle exoskeleton through the flexible Bowden cable.

B. Torque Control

Assistive torque was tracked using a combination of proportional-derivative control and iterative learning algorithm (PD-LRN) [25], [26]. Iterative learning control is suitable for systems performed the same tasks repetitively and periodically [27]. Human walking exhibits cyclic behavior. When human walking assisted by an ankle exoskeleton on a treadmill, torque error in the last gait cycle can be used to modify motor velocity for the next gait cycle. A P-type iterative learning controller was used to reduce the steady state errors of torque tracking. The PD-LRN torque controller can be expressed by the following equation [25], [26]:

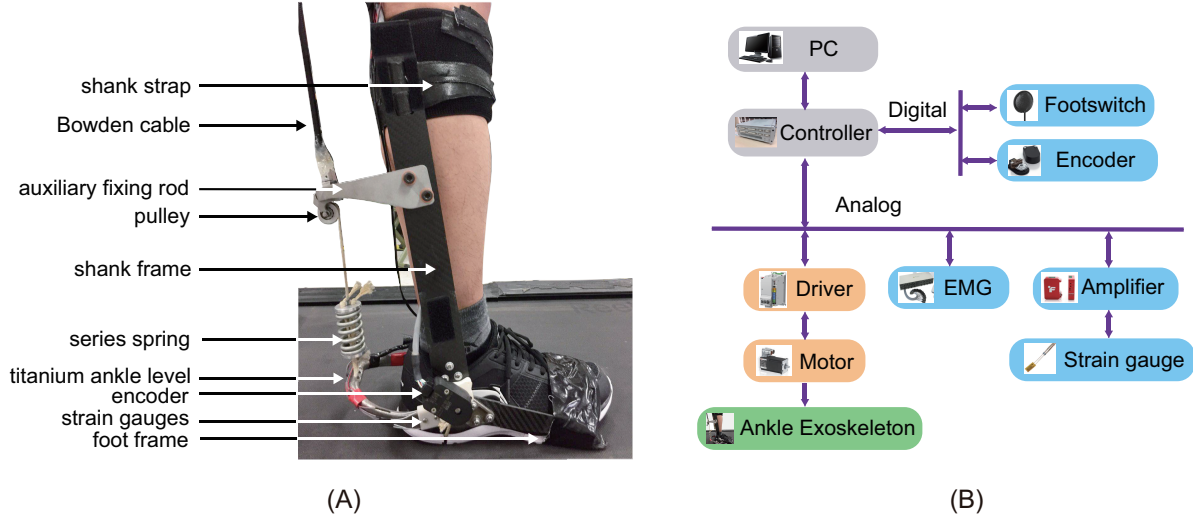


Fig. 2. **Ankle exoskeleton mechanical structure and system hardware configuration.** (A) Mechanical structure of the ankle exoskeleton. Shank frame, foot frame and titanium ankle level joined together in biological ankle joint. A Bowden cable was threaded a pulley and a series spring, and connected with the ankle level. A Wheatstone-bridge consisted of four strain gauges was used to measure the assistive torque. An encoder was installed on the exoskeleton joint shaft to measure the ankle joint angle. A footswitch was attached inside the exoskeleton shoe to detect heel strike. (B) Hardware configuration of the system. The system comprised: (1) an ankle exoskeleton (green), (2) a high-speed control system (grey), (3) sensors that measuring muscles EMG signals and exoskeleton performance (blue), and (4) motor and driver (orange).

TABLE I
PARAMETER VALUES OF PD-LRN TORQUE CONTROLLER.

K_P	K_D	K_L	K_{flt}	β	D	T
2.2	0.01	0.15	1	1	0.018s	0.050s

$$\begin{aligned}
 \dot{\theta}_{m,des}(i, n) &= \frac{1}{T} \cdot \Delta\theta_{m,des}(i, n) \\
 \Delta\theta_{m,des}(i, n) &= -K_P \cdot e_\tau(i, n) - K_D \cdot \dot{\theta}_m(i, n) \\
 &\quad + \Delta\theta_{m,des}^{LRN}(i + D, n) \\
 \Delta\theta_{m,des}^{LRN}(i, n + 1) &= \beta \cdot \theta_{m,des}^{LRN}(i, n) - K_L \cdot e_{flt}(i, n) \\
 e_{flt}(i, n) &= (1 - K_{flt}) \cdot e_{flt}(i, n - 1) \\
 &\quad + K_{flt} \cdot e_\tau(i, n)
 \end{aligned} \tag{1}$$

where $\dot{\theta}_{m,des}$ is desired motor velocity, $\Delta\theta_{m,des}$ is the desired motor displacement. $\dot{\theta}_m$ is measured motor velocity. $\Delta\theta_{m,des}^{LRN}$ is learned feed-forward compensation from iterative learning algorithm, and e_{flt} is a torque error filter. K_P , K_D and K_L are proportional gain, derivative gain and iterative learning gain. $e_\tau = \tau - \tau_{des}$ is torque error. i is the time index within a gait cycle, n is the stride number. T is a constant related to the motor rise time. D is the estimated number of time indices for the time delay between motor velocity commands and motor rotation. The time delay was about 18 ms according to our previous tests. $K_{flt} \in [0, 1]$ is a weight on the learned error. $\beta \in [0, 1]$ is a weighting term of the learned torque trajectory. Torque control parameters were systematically adjusted to minimize the tracking error (Table I). The reported tracking error was as low as 1% of peak desired torque. No sensible delay or oscillation was detected by the participants [25].

C. Assistive Torque Profile

We defined the desired plantarflexion assistive torque profile as a single-peak curve according to human walking biomechanics [28], [29]. We used two cubic splines joined at their peaks to realize this curve [8]. Then two parameters were used to define the assistive torque profile: peak time and peak torque (Fig. 3(A)). Peak time was referred to the time that ankle exoskeleton provided the maximum plantarflexion assistive torque. Peak time was normalized to percent stride and was denoted as X%. Peak torque was referred to the maximum plantarflexion assistive torque that ankle exoskeleton provided for the wearer. Peak torque was normalized to body weight and was presented in unit of $\text{N}\cdot\text{m}\cdot\text{kg}^{-1}$.

To compare the lower leg muscle activities of ankle exoskeleton assisted walking, we designed 9 different exoskeleton assistance conditions that combined three different peak times and three different peak torque levels (Fig. 3(B)). Three peak times were 46%, 49% and 52% of a gait cycle (represented as early, middle, and late peak time). We constrained peak time to between 45% and 53% of a gait cycle because most healthy subjects ankle push-off timings were in this range [22], [28], [29]. Moreover, these three peak time values that we selected could distinguish different assistance and were to be close to the optimal peak time from a previous study [8]. Three plantarflexion peak torque levels were 0.3, 0.5 and 0.7 $\text{N}\cdot\text{m}\cdot\text{kg}^{-1}$ (represented as low, medium, and high peak torque). Peak torque was bounded within 0.3 to 0.7 $\text{N}\cdot\text{m}\cdot\text{kg}^{-1}$ to ensure comfortability during walking test. Too higher assistive peak torque could induce discomfort in cases with unusual timing. Too smaller assistive peak torque would be unable to provide appreciable assistance. The nine different assistance conditions were thus referred as Early-time-Low-torque (EL), Early-time-Medium-torque (EM), Early-time-High-torque (EH), Middle-time-Low-torque (ML), Middle-time-Medium-torque (MM),

Middle-time-High-torque (MH), Late-time-Low-torque (LL), Late-time-Medium-torque (LM) and Late-time-High-torque (LH) respectively.

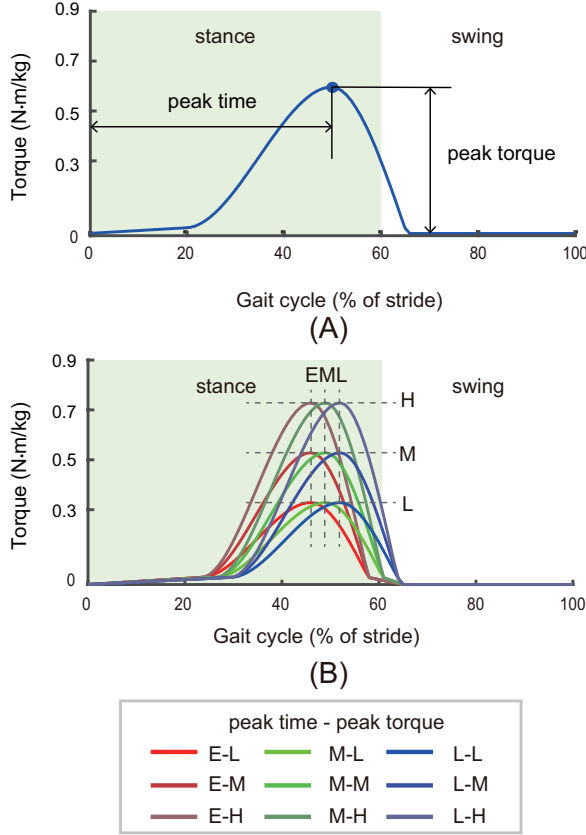


Fig. 3. **Plantarflexion assistive torque profile.** (A) The assistive torque profile. Peak time referred to the time that ankle exoskeleton provided the maximum ankle plantarflexion torque. Peak torque referred to the maximum assistive torque that the exoskeleton provided for the wearer. (B) Nine different exoskeleton assistance conditions that applied in the experiment. Based on the combinations of three peak times and three peak torque levels, the 9 different assistance conditions were Early-time-Low-torque (EL), Early-time-Medium-torque (EM), Early-time-High-torque (EH), Middle-time-Low-torque (ML), Middle-time-Medium-torque (MM), Middle-time-High-torque (MH), Late-time-Low-torque (LL), Late-time-Medium-torque (LM) and Late-time-High-torque (LH). We grouped exoskeleton assistance conditions by peak time. Early, Middle and Late peak time are marked in red, green and blue respectively. Darker colors refer to higher peak torque levels. The light green rectangle shows the stance phase in a gait cycle. Dash lines show the three peak times and three peak torque levels.

D. Participants and Experimental Protocol

Nine healthy male subjects (age 22.3 ± 3.2 years; weight 67.4 ± 9.5 kg; height 177.1 ± 4.2 cm; mean $\pm$ S.D.) participated in the experiment. All participants were provided with written informed consent before completing the protocol, which was approved by the ethical committee of the Nankai University.

The experimental protocol included two sessions for each participants: a training session and a testing session. Participants walked on a treadmill at $1.25 \text{ m}\cdot\text{s}^{-1}$ while wearing the ankle exoskeleton on right ankle joint in all sessions.

The training session was performed on the first day. All exoskeleton assistance conditions were applied to the subjects

to ensure that they adapt to walking with the exoskeleton. Moreover, subjects were advised to relax lower leg muscles and not to resist the exoskeleton assistance.

The testing session was performed on a separate day, at least 48 hours later to avoid fatigue-induced effects. It started with a 4 minutes warming-up and followed by a 2 minutes normal walking (NW) condition in which participants walked without the exoskeleton. Next, subjects performed a 2 minutes zero-torque (ZT) condition in which they walked with the exoskeleton applying zero assistive torque to the human ankle. Then the 9 different assistance conditions were applied in random order. To minimize the effects of adaptation and fatigue, subjects were asked to experience the 9 different assistance conditions again but in a reverse order. Then the testing session ended by subjects performing a ZT condition. All conditions lasted 2 minutes to ensure that muscle activities reached steady-state. Resting breaks of 2 to 5 minutes were given between all conditions according to subjects' own requests. One subject did not test the conditions in reverse order due to hardware malfunction. Another two subjects unfinished LH condition because they can't adapt to this condition.

E. Data Collection and Analysis

We measured lower leg muscle activities using a wired EMG system. Surface electromyography signals were measured from 5 right lower leg muscles: tibialis anterior (TA), medial soleus (m.SOL), lateral soleus (l.SOL), medial gastrocnemius (m.GAS) and lateral gastrocnemius (l.GAS) (Fig. 4). These muscles are primarily responsible for human ankle plantarflexion and dorsiflexion movements in sagittal plane. Electrodes were placed according to the guidance in [30]. Sampling frequency were set to 5000 Hz. Raw EMG signals were high-pass filtered with a 2nd order Butterworth filter (cut-off frequency 20 Hz), full-wave rectified, and low-pass filtered with a 2nd order Butterworth filter (cut-off frequency 10 Hz). All preprocessed EMG amplitudes were then normalized to the average EMG peak value during the last minute of the normal walking session for the same participants.

Three electromyography-based metrics in time domain were calculated to quantify changes of muscle activities during exoskeleton-assisted walking: root-mean-square (RMS) value, mean value and peak value of measured lower leg muscle electromyography over a whole stride [31], [32]. EMG metrics for each condition were calculated from the last minute of corresponding walking data. Pearson correlation coefficients was calculated to evaluate the correlation between all electromyography metrics.

F. Statistical Analysis

Repeated measures analyses of variance (ANOVA) including 10 conditions (NW, EL, ML, LL, EM, MM, LM, EH, MH, LH) were conducted to verify the effects of plantarflexion assistance on lower leg muscle activities compared to NW condition. Then repeated measures ANOVA including 10 conditions (ZT, EL, ML, LL, EM, MM, LM, EH, MH, LH) were conducted to identify differences in muscle activities between all assistance conditions and ZT condition. The significance

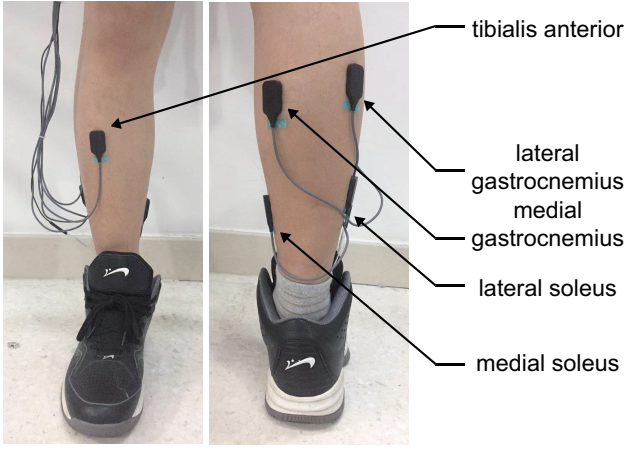


Fig. 4. **Electromyography electrodes in the right lower leg muscles.** Electromyography signals were measured from five lower leg muscle positions: tibialis anterior, medial soleus, lateral soleus, medial gastrocnemius and lateral gastrocnemius. Electrodes were fixed by medical adhesive tapes to keep in close contact with skin.

TABLE II
PEARSON CORRELATION COEFFICIENTS ACROSS DIFFERENT EMG METRICS.

R	mean value	max value	RMS value
mean value	-	75.5%	90.0%
max value	75.5%	-	95.4%
RMS value	90.0%	95.4%	-

level was set at $\alpha = 0.05$. Statistical analyses were conducted with Matlab (MathWorks, USA). We also performed student's paired t -test to compare RMS of lateral soleus muscle activity of 9 exoskeleton assistance conditions with each other.

III. RESULTS

Table II shows the Pearson correlation coefficients between all electromyography metrics. Lower leg muscle EMG RMS values were well corrected with both mean and max values, with correlation coefficients being 90.0% and 95.4% respectively. Therefore, muscle activities RMS values were selected as the primary metric to be investigated.

The lateral soleus muscle activity, particularly in stance phase, was reduced most significantly during human walking assisted by the ankle exoskeleton (Fig. 5(A), Fig. 6). Under one specific assistance condition, middle peak time and high peak torque (MH), the RMS of lateral soleus muscle activity was 0.195 ± 0.042 . This means a reduction of $33.6 \pm 14.0\%$ compared to NW condition ($P < 0.05$), and a reduction of $24.9 \pm 19.5\%$ compared to ZT condition ($P < 0.05$). Fairly large differences in RMS of lateral soleus muscle activity were found under this condition with respect to other 6 assistance conditions (EL, $P = 5.5 \cdot 10^{-3}$; EM, $P = 1.5 \cdot 10^{-2}$; EH, $P = 1.3 \cdot 10^{-3}$; ML, $P = 4.3 \cdot 10^{-3}$; MM, $P = 7.0 \cdot 10^{-3}$; LL, $P = 1.8 \cdot 10^{-2}$). No significant differences were found between MH and other 2 assistance conditions (LM, $P = 0.066$; LH, $P = 0.73$). Furthermore, under LH and MM assistance condition, the RMS value of lateral soleus activity significantly decreased

compared to walking with the exoskeleton but provide zero assistive torque ($P = 3.7 \cdot 10^{-2}$, $P = 9.0 \cdot 10^{-3}$, Fig. 6).

There were no significant differences in the tibialis anterior muscle activity between exoskeleton assistance conditions and normal walking condition ($P = 0.71$, Fig. 5(A)). For the medial soleus, changes of muscle activity under different assistance conditions were not obvious compared to normal walking condition ($P = 0.96$, Fig. 5(A)). For the medial and lateral gastrocnemius, changes of muscle activities were not obvious compared to normal walking condition ($P = 1.0$, $P = 1.0$, Fig. 5(A)). No significant changes of these muscle activities under exoskeleton assisted walking conditions compared to zero-torque condition.

IV. DISCUSSION

The aim of this research was to evaluate lower leg muscle activities during human walking assisted by an ankle exoskeleton. In our experiments, the assistive torque was applied accurately to the wearers by using a PD-LRN control algorithm. We designed nine exoskeleton assistance conditions to evaluate lower leg muscle activity changes. Meanwhile, the RMS values of the measured EMG signals from right lower leg were used as an indicator of EMG-based metric.

The lateral soleus muscle activity showed the biggest differences among different walking conditions. It had decreased activities under all assistance conditions compared to NW and ZT condition. Meanwhile, the reductions of lateral soleus activity varied among different assistance conditions. This result indicates that ankle exoskeleton assistance has a positive effect on lateral soleus, and also suggests that the lateral soleus muscle activity is suitable for being selected as a candidate objective of HIL optimization. The significant reductions of the lateral soleus muscle activity during exoskeleton assistance can be explained from biomechanics. The soleus muscle connects calf and phalanges, and has a higher proportion of slow muscle fibers. Soleus muscle contractions complete the primary plantarflexion movements during human walking [33]. Therefore, when plantarflexion assistive torque was applied to the wearer through the exoskeleton, the soleus muscle contractions were partially replaced by the exoskeleton assistance, led to soleus muscle activity reductions. This explanation also suggests that the assistive torque that our ankle exoskeleton provided was consistent with human walking biomechanics.

Changes of other lower leg muscles during ankle exoskeleton assisted walking were not significant. The tibialis anterior muscle activity had no obvious changes when assisted by low and medium peak torque. However, muscle activity increased when applied high peak torque, Especially at early and middle peak time (Fig. 5(B)). This may be due to the coactivation of the tibialis anterior to counteract inappropriate exoskeleton assistive torque. From biomechanics, TA acts antagonistically to the exoskeleton which means high peak torque may negatively influence the muscle activity. This also suggests that TA can be used as a balance term in HIL optimization objective, and preferable assistance seems to be a trade-off between more lateral soleus activity reductions and less tibialis anterior activity increases. For the m.SOL, except under LH condition,

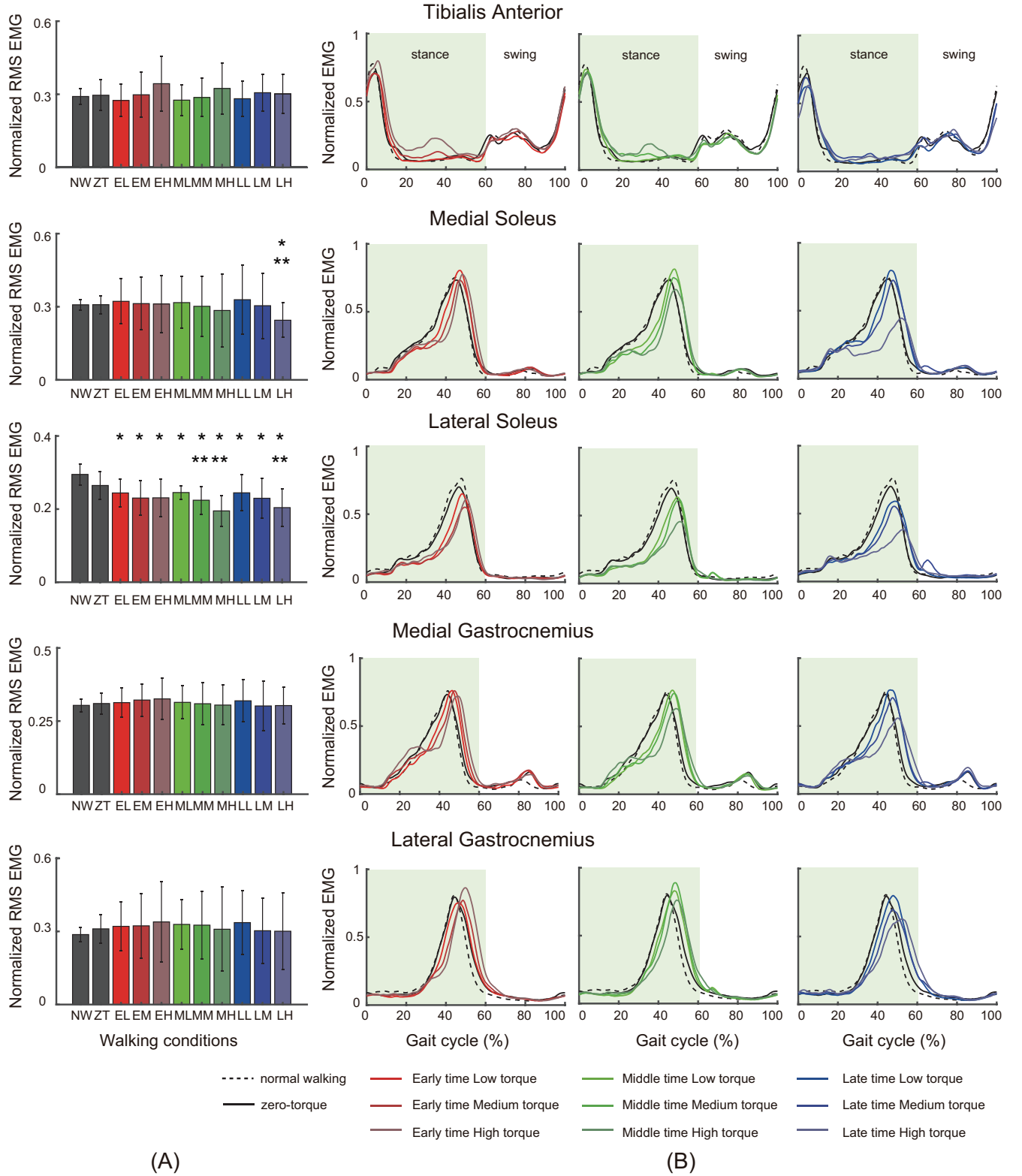


Fig. 5. **Lower leg muscle activities over the whole stride in different exoskeleton assistance conditions.** (A) RMS normalized stride-wise electromyography from $N = 9$ subjects. Bars and whiskers are subject means and standard deviations of the whole stride RMS average muscle activity. * represent statistically significant differences with respect to normal walking condition. ** represent statistically significant differences with respect to zero-torque condition. (B) Lower leg muscle activities (tibialis anterior, medial soleus, lateral soleus, medial gastrocnemius and lateral gastrocnemius) over the stride from one side heel contact (0% of a gait cycle) to opposite side toe off (100% of a gait cycle) in human walking in different ankle exoskeleton assistance conditions. We grouped exoskeleton assistance conditions by peak time. Early, Middle and Late peak time are marked in red, green and blue respectively. Darker colors refer to higher peak torque levels. The light green rectangle shows the stance phase in a gait cycle.

TABLE III
RMS OF LOWER LEG MUSCLE ACTIVITIES DURING DIFFERENT WALKING CONDITIONS.

Peak Time (% of stride)	Peak Torque (N·m·kg ⁻¹)	RMS				
		TA	m.SOL	l.SOL	m.GAS	l.GAS
46	0.3	0.274±0.066	0.322±0.092	0.244±0.037*	0.313±0.051	0.321±0.10
	0.5	0.298±0.093	0.313±0.108	0.230±0.047*	0.322±0.055	0.323±0.132
	0.7	0.344±0.113	0.311±0.118	0.231±0.051*	0.326±0.071	0.339±0.165
49	0.3	0.276±0.062	0.317±0.106	0.245±0.019*	0.314±0.056	0.329±0.101
	0.5	0.287±0.079	0.302±0.123	0.224±0.038*,**	0.309±0.073	0.326±0.140
	0.7	0.324±0.104	0.285±0.149	0.195±0.042*,**	0.305±0.068	0.309±0.172
52	0.3	0.282±0.071	0.329±0.142	0.245±0.049*	0.319±0.073	0.336±0.130
	0.5	0.307±0.075	0.304±0.134	0.230±0.054*	0.302±0.085	0.303±0.134
	0.7	0.302±0.079	0.245±0.071*,**	0.204±0.051*,**	0.303±0.062	0.301±0.158
normal walking		0.291±0.031	0.308±0.022	0.295±0.028	0.304±0.022	0.287±0.029
zero-torque		0.296±0.063	0.309±0.037	0.265±0.037	0.310±0.036	0.311±0.060

* statistically significant differences with respect to normal walking condition

** statistically significant differences with respect to zero-torque condition

TA, tibialis anterior; m.SOL, medial soleus; l.SOL, lateral soleus; m.GAS, medial gastrocnemius; l.GAS, lateral gastrocnemius

muscle activity had no statistically significant differences with respect to NW condition or ZT condition. The same as l.SOL, m.SOL activity significantly decreased under LH condition compared to NW and ZT condition ($P < 0.05$, $P < 0.05$). This suggests that assistance with late peak time and high peak torque may further decrease lower leg muscle activities during exoskeleton assisted walking. The m.GAS and l.GAS muscle activity under all exoskeleton assistance conditions were not significantly changed compared to NW condition and ZT condition. Peak EMG changes were not observed compared to NW condition (Fig. 5(B)). From biomechanics aspects, the gastrocnemius has higher proportion of fast muscle fibers, and is primarily involved in movements like running and jumping that need to provide greater muscle contraction force in a short period of time. However, the gastrocnemius is less involved in walking and standing. This explanation suggests that the gastrocnemius may be not a good choice for evaluating ankle exoskeleton assistance performance. Moreover, for our ankle exoskeleton, the shank strap of the ankle exoskeleton is close to the gastrocnemius. This may affect gastrocnemius contraction and increased measuring noise of EMG signals around strap.

We also analyzed the effects of peak time and peak torque on lateral soleus activity independently. There were no significant differences in lateral soleus activity between early, middle and late peak time assistance conditions. For example, under EM, MM and LM conditions, the RMS values of lateral soleus activity have no obvious changes. This suggests that peak time may be not a determinant factor of the exoskeleton assistance that can influence lateral soleus activity. However, significant differences were found in lateral soleus activity between low, medium and high peak torque assistance conditions ($P < 0.05$). As shown in Fig. 6, higher peak torque could reduce more lateral soleus activity. This indicates that exoskeleton assistive peak torque levels can greatly affect lateral soleus activity. This is also consistent with our expectations that human biological efforts like muscle activity will decrease as the assistive torque levels increases.

It is worth noting that our experiments have limitations.

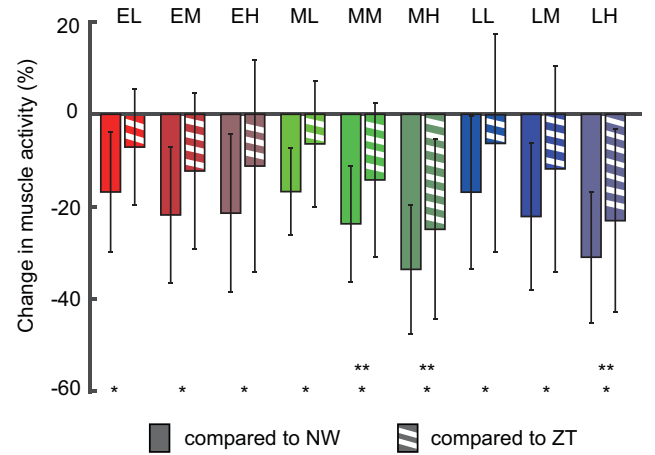


Fig. 6. Changes of lateral soleus activity in different exoskeleton assistance conditions with respect to normal walking condition and zero-torque condition. Muscle activity was quantified as the RMS of the processed EMG signals. Bars and whiskers are subject means and standard deviations of the whole stride RMS average lateral soleus muscle activity. Solid bars and striped bars indicate change of muscle activity in exoskeleton assistance conditions compared to normal walking condition and zero-torque condition. We grouped exoskeleton assistance conditions by peak time. Early, Middle and Late peak time are marked in red, green and blue respectively. Darker colors refer to higher peak torque levels. EL: Early-time-Low-torque; EM: Early-time-Medium-torque; EH: Early-time-High-torque; ML: Middle-time-Low-torque; MM: Middle-time-Medium-torque; MH: Middle-time-High-torque; LL: Late-time-Low-torque; LM: Late-time-Medium-torque; LH: Late-time-High-torque; NW: normal walking; ZT: zero-torque. * represent statistically significant differences with respect to normal walking condition. ** represent statistically significant differences with respect to zero-torque condition.

One is that the participants we currently recruited were almost young people. Both old subjects and disabled subjects would be recruited in future works. Another is that we conducted a small scale test to compare metabolic cost and muscle activities under MH, EH, NW and ZT condition. According to the one subject data, the lateral soleus activity decreased under MH and EH condition. Metabolic cost decreased in this two conditions compared to ZT condition, and it increased compared to NW condition. The relationship between muscle activities and metabolic cost under exoskeleton assisted walk-

ing seems to be complicated. Future works would be required to recruit more subjects and to conduct more experiments to explore the relationship among metabolic cost, muscle activities and other physiological responses under different exoskeleton assistance conditions.

V. CONCLUSIONS

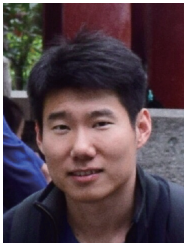
In this paper, we systematically evaluated lower leg muscle activities and quantified changes of electromyography-based metrics during human walking assisted by an ankle exoskeleton. The lateral soleus muscle activity showed significant reductions when exoskeleton assistance was applied, and it was most reduced under one of the nine conditions. The lateral soleus muscle activity has the potential to be used as a candidate objective of HIL optimization. In future works, we plan to recruit different ages subjects and to conduct experiments to compare the relationship between muscle activities and other physiological responses during different exoskeleton assistance conditions. We also plan to conduct experiments to explore whether muscle activity can be used as a physiological objective used in human-in-the-loop optimization of exoskeleton assistance.

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REFERENCES

- [1] W. Cornwall, "In pursuit of the perfect power suit," *Science*, vol. 350, no. 6258, pp. 270–273, 2015.
- [2] A. M. Dollar and H. Herr, "Lower extremity exoskeletons and active orthoses: Challenges and state-of-the-art," *IEEE Trans. Robot.*, vol. 24, no. 1, pp. 144–158, 2008.
- [3] A. J. Young and D. P. Ferris, "State of the art and future directions for lower limb robotic exoskeletons," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 2, pp. 171–182, 2017.
- [4] P. Malcolm, W. Derave, S. Galle, and D. D. Clercq, "A simple exoskeleton that assists plantarflexion can reduce the metabolic cost of human walking," *PLoS ONE*, vol. 8, 2013.
- [5] G. Lee, J. Kim, F. A. Panizzolo, Y. M. Zhou, L. M. Baker, I. Galiana, P. Malcolm, and C. J. Walsh, "Reducing the metabolic cost of running with a tethered soft exosuit," *Sci. Robot.*, vol. 2, 2017.
- [6] M. Chandrapal, X. Chen, and W. Wang, "Preliminary evaluation of a lower-limb exoskeleton - star climbing," in *IEEE/ASME Int. Conf. Adv. Intel. Mechatron.*, 2013, pp. 1458–1463.
- [7] J. R. Koller, D. H. Gates, D. P. Ferris, and C. D. Remy, "'body-in-the-loop' optimization of assistive robotic devices: A validation study," *Robotics: Science and Systems*, 2016.
- [8] J. Zhang, P. Fiers, K. A. Witte, R. W. Jackson, K. L. Poggensee, C. G. Atkeson, and S. H. Collins, "Human-in-the-loop optimization of exoskeleton assistance during walking," *Science*, vol. 356, pp. 1280–1284, 2017.
- [9] Y. Ding, M. Kim, S. Kuindersma, and C. J. Walsh, "Human-in-the-loop optimization of hip assistance with a soft exosuit during walking," *Sci. Robot.*, vol. 3, 2018.
- [10] L. Schiatti, J. Tessadori, N. Deshpande, G. Barresi, L. C. King, and L. S. Mattos, "Human in the loop of robot learning: Eeg-based reward signal for target identification and reaching task," in *Proc. IEEE Int. Conf. Robot. Autom.*, vol. 3, 2018, pp. 4473–4480.
- [11] A. Gams, T. Petric, T. Debevec, and J. Babic, "Effects of robotic knee exoskeleton on human energy expenditure," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 6, pp. 1636–1644, 2013.
- [12] S. Galle, P. Malcolm, S. H. Collins, and D. D. Clercq, "Reducing the metabolic cost of walking with an ankle exoskeleton: interaction between actuation timing and power," *J. Neuroeng. Rehabil.*, vol. 14, no. 35, 2017.
- [13] Y. Ding, F. A. Panizzolo, C. Sivi, P. Malcolm, I. Galiana, K. G. Holt, and C. J. Walsh, "Effect of timing of hip extension assistance during loaded walking with a soft exosuit," *J. Neuroeng. Rehabil.*, vol. 13, no. 87, 2016.
- [14] P. Malcolm, R. E. Quesada, J. M. Caputo, and S. H. Collins, "The influence of push-off timing in a robotic ankle-foot prosthesis on the energetics and mechanics of walking," *J. Neuroeng. Rehabil.*, vol. 12, no. 21, 2015.
- [15] S. H. Collins, M. B. Wiggin, and G. S. Sawicki, "Reducing the energy cost of human walking using an unpowered exoskeleton," *Nature*, vol. 522, pp. 212–215, 2015.
- [16] J. R. Koller, D. A. Jacobs, D. P. Ferris, and C. D. Remy, "Learning to walk with an adaptive gain proportional myoelectric controller for a robotic ankle exoskeleton," *J. Neuroeng. Rehabil.*, vol. 12, no. 97, 2015.
- [17] O. M. Blake and J. M. Wakeling, "Estimating changes in metabolic power from emg," *Springerplus*, vol. 2, no. 229, 2013.
- [18] C. Fleischer and G. Hommel, "A human-exoskeleton interface utilizing electromyography," *IEEE Trans. Robot.*, vol. 24, no. 4, pp. 872–882, 2008.
- [19] P. G. Jung, G. Lim, S. Kim, and K. Kong, "A wearable gesture recognition device for detecting muscular activities based on air-pressure sensors," *IEEE Trans. Ind. Inform.*, vol. 11, no. 2, pp. 485–494, 2015.
- [20] Y. Zhuang, S. Yao, C. Ma, and R. Song, "Admittance control based on emg-driven musculoskeletal model improves the human-robot synchronization," *IEEE Trans. Ind. Inform.*, vol. 15, no. 2, pp. 1211–1218, 2019.
- [21] K. E. Gordon and D. P. Ferris, "Learning to walk with a robotic ankle exoskeleton," *J. Biomech.*, vol. 40, no. 12, pp. 2636–2644, 2007.
- [22] R. W. Jackson and S. H. Collins, "An experimental comparison of the relative benefits of work and torque assistance in ankle exoskeletons," *J. Appl. Physiol.*, vol. 119, pp. 541–557, 2015.
- [23] K. A. Witte, J. Zhang, R. W. Jackson, and S. H. Collins, "Design of two lightweight, high-bandwidth torque-controlled ankle exoskeletons," in *IEEE Int. Conf. Robot. Autom.*, 2015, pp. 1223–1228.
- [24] J. F. Veneman, R. Ekkelenkamp, R. Kruidhof, F. C. T. V. D. Helm, and H. V. D. Kooij, "Design of a series elastic- and bowden cable-based actuation system for use as torque-actuator in exoskeleton-type training," in *Proc. 9th IEEE Int. Conf. Rehabil. Robot.*, 2005, pp. 496–499.
- [25] J. Zhang, C. C. Cheah, and S. H. Collins, "Experimental comparison of torque control methods on an ankle exoskeleton during human walking," in *IEEE Int. Conf. Robot. Autom.*, 2015, pp. 5584–5589.
- [26] J. Zhang, C. C. Cheah, and S. H. Collins, "Torque control in legged locomotion," in *Bioinspired Legged Locomotion: Models, Concepts, Control and Application*, 2017, pp. 375–428.
- [27] P. G. Jung, G. Lim, S. Kim, and K. Kong, "A survey of iterative learning control," *J. IEEE Contr. Syst.*, vol. 26, no. 3, pp. 0–114, 2006.
- [28] Y. L. Han and X. S. Wang, "The biomechanical study of lower limb during human walking," *Sci. China Technol. Sc.*, vol. 54, no. 4, pp. 983–991, 2011.
- [29] M. W. Whittle, Ed., *An Introduction to Gait Analysis*, 4th ed. Oxford: Butterworth-Heinemann, 2004.
- [30] H. J. Hermens, B. Freriks, C. D. Klug, and G. Rau, "Development of recommendations for semg sensors and sensor placement procedures," *J. Electromyogr. Kinesiol.*, vol. 10, no. 5, pp. 361–374, 2000.
- [31] A. Phinyomark, C. Limsakul, and P. Phukpattaranont, "Emg feature extraction for tolerance of 50 hz interference," in *Proc. 4th PSU-UNS Int. Conf. Eng. Tech.*, 2009, pp. 289–293.
- [32] R. Boostani and M. H. Moradi, "Evaluation of the forearm emg signal features for the control of a prosthetic hand," *J. Physiol. Meas.*, vol. 24, no. 2, pp. 309–319, 2003.
- [33] F. C. Anderson and M. G. Pandy, "Individual muscle contributions to support in normal walking," *Gait Posture*, vol. 17, pp. 159–169, 2003.



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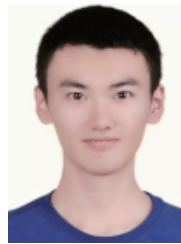
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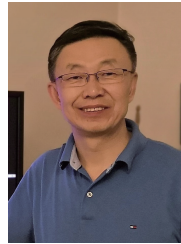
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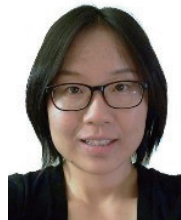
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